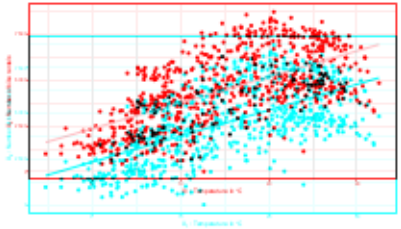
GENERALIZED ADDITIVE MODEL (GAM)

- Hastie and Tibshirani (1986)

Problem: LM not great if features act on outcome non-linearly target variable is not linear



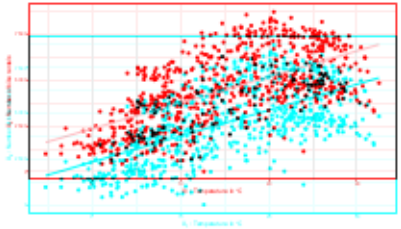
GENERALIZED ADDITIVE MODEL (GAM)

▸ Hastie and Tibshirani (1986)

Problem: LM not great if features act on outcome non-linearly / relationship between features and target variable is not linear

Workaround in LMs / GLMs:

- Feature transformations (e.g., exp or log)
- Including high-order effects
- Categorization of features (i.e., intervals/ buckets of feature values)



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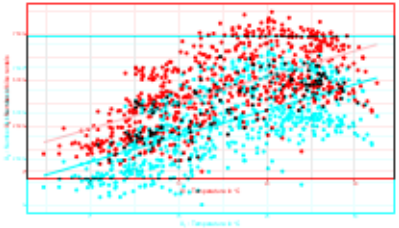


Idea of GAMs:

- Instead of linear terms $\theta_j x_j$, use flexible functions $f_j(x_j)$ $\rightsquigarrow$ splines

$$g(\mathbb{E}(y \mid \mathbf{x})) = \theta_0 + f_1(x_1) + f_2(x_2) + \ldots + f_p(x_p)$$

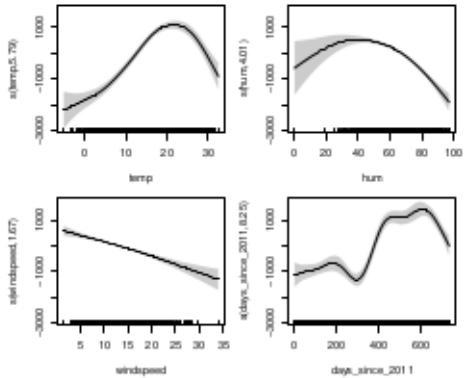
- Preserves additive structure and allows to model non-linear effects
- Splines have a smoothness parameter to control flexibility (prevent overfitting)
  $\rightsquigarrow$ Needs to be chosen, e.g., via cross-validation

GENERALIZED ADDITIVE MODEL (GAM) - EXAMPLE

Fit a GAM with smooth splines for four numeric features of bike rental data

⇝ more flexible and better model fit but less interpretable than LM

	edf	p-value
s(temp)	5.8	0.00
s(hum)	4.0	0.00
s(windspeed)	1.7	0.00
s(days_since_2011)	8.3	0.00

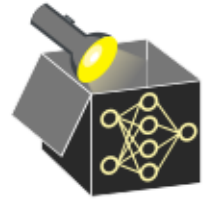


Interpretation

- Interpretation needs to be done visually and relative to average prediction, also see PDPs
- Edf (effective degrees of freedom) represents complexity of smoothness

MODEL-BASED BOOSTING

Bühlmann and Yu 2003



- Boosting iteratively combines weak base learners to create powerful ensemble
- Idea: Use simple BLs (e.g univariate, with splines) to ensure interpretability
- Possible to combine BL of same type (with distinct parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^*$):

$$b^{[j]}(\mathbf{x}, \boldsymbol{\theta}) + b^{[j]}(\mathbf{x}, \boldsymbol{\theta}^*) = b^{[j]}(\mathbf{x}, \boldsymbol{\theta} + \boldsymbol{\theta}^*)$$

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Bühlmann and Yu 2003

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$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) \equiv b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

- In each iteration, fit a set of BLs, add best one to model (with step-size ν):

$$\begin{aligned}
\hat{f}^{[1]} &= \hat{f}_0 + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]}) \\
\hat{f}^{[2]} &= \hat{f}^{[1]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[2]}) \\
\hat{f}^{[3]} &= \hat{f}^{[2]} + \nu b^{[1]}(\mathbf{x}_1, \theta^{[3]}) \\
&= \hat{f}_0 + \nu \left(b^{[3]}(\mathbf{x}_3, \theta^{[1]} + \theta^{[2]}) + b^{[1]}(\mathbf{x}_1, \theta^{[3]}) \right) \\
&= \hat{f}_0 + \hat{f}_3(\mathbf{x}_3) + \hat{f}_1(\mathbf{x}_1)
\end{aligned}$$

- Final model is additive GAM, we can read off effect curves

MODEL-BASED BOOSTING - LINEAR EXAMPLE

Simple case: Use linear model with single feature (including intercept) as BL

- Idea: Use simple linear BL to ensure interpretability (in general also spline BL possible)

$$b^{[j]}(x, \theta) = x_j \theta + \theta_0 \quad \text{for } j = 1, \dots, p \quad \rightsquigarrow \text{ordinary linear regression}$$

- Possible to combine linear BL of same type (with distinct parameters θ and θ^*):

$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) = b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

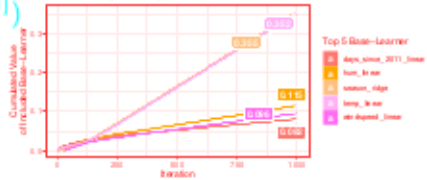
- Here: Interpretation of weights as in LM
- After many iterations, it converges to same solution as LM
- In each iteration, fit a set of BLs and add the best BL to previous model (using step-size ν):

$$\hat{f}^{[m]} = \hat{f}^{[m-1]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]})$$

1000 iter. with $\nu = 0.1$	Intercept	Weights
days_since_2011	-1791.06	4.9
hum	1953.05	-31.1
season	0	WINTER: -323.4 SPRING: 539.5 SUMMER: -280.2 FALL: 67.2
temp	-1839.85	120.4
windspeed	725.70	-56.9
offset	4504.35	

$\Rightarrow$ Converges to solution of LM

Relative frequency of selected BLs across iterations



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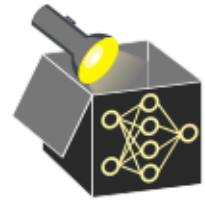

Simple case: Use linear model with single feature (including intercept) as BL

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$$b^{[j]}(x, \theta) = x_j\theta + \theta_0 \quad \text{for } j = 1, \dots p \quad \rightsquigarrow \text{ordinary linear regression}$$

- Possible to combine linear BL of same type (with distinct parameters θ and θ^*):

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- Here: Interpretation of weights as in LM
- After many iterations, it converges to same solution as LM
- Early stopping allows feature selection & may prevent overfitting (regularization)

$$\hat{f}^{[1]} = \hat{f}^{[0]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]})$$

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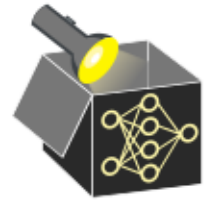
offset	4504.35	

$\Rightarrow$ Converges to solution of LM

20 iter. with $\nu = 0.1$	Intercept	Weights
days_since_2011	-1210.27	3.3
season	0	WINTER: -276.9 SPRING: 137.6 SUMMER: 112.8 FALL: 20.3
temp	-1118.94	73.2
offset	4504.35	

$\Rightarrow$ 3 BLs selected after 20 iter. (feature selection)

LINEAR EXAMPLE: INTERPRETATION



- Recall: Boosting iteratively combines weak base learners (BL)
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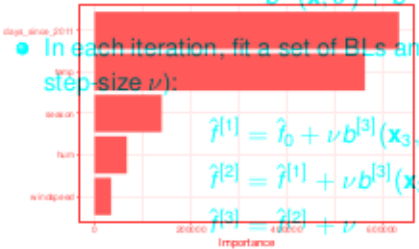
$$= \hat{f}_0 + \hat{f}_3(\mathbf{x}_3) +$$

Feature importance: aggregated change in risk in each iteration per feature

- E.g. iteration 1: days_since_2011 with risk reduction (MSE) of 140,782.94
- For every iteration the change in risk can be attributed to a feature
- Possible to combine linear BL of same type (with distinct parameters θ and θ^*):

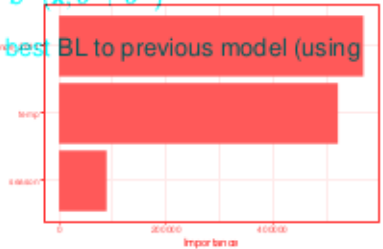
$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) = b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

1000 iterations



In-bag-risk: 434,686.0
OOB risk (10-fold CV): 446,450.0

20 iterations



In-bag-risk: 693,505.0
OOB risk (10-fold CV): 705,776.0

$\Rightarrow$ Difference in risk: 258,819.0
Difference in OOB risk: 259,326.0

NON-LINEAR EXAMPLE: INTERPRETATION

- Fit model on bike data with different BL types (1000 iter.) Daniel Schalk et al. 2018
- BLs: linear and centered splines for numeric features, categorical for season
- Idea: Use simple linear BL to ensure interpretability (in general also spline BL possible)
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\end{aligned}$$

- Final model is additive (as GAMs), where each component function is interpretable

NON-LINEAR EXAMPLE: INTERPRETATION

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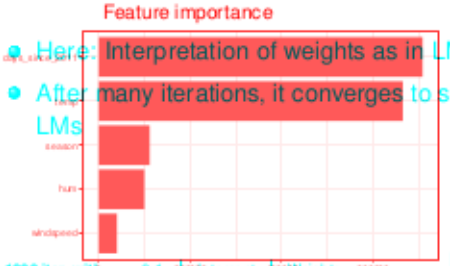

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- After many iterations, it converges to same solution as least square estimate of LMs

1000 iter. with $\nu = 0.1$

	Intercept	Weights
days_since_2011	-1791.06	4.9
season	0	SPRING: 539.5 SUMMER: 989.2 FALL: 87.2
windspeed	725.70	-56.9

Relating frequency of selected BLs across iterations

⇒ In-bag-risk: 250,202.0 ; OOB risk (10-fold CV): 267,497.0 (difference to lin. example: 178,953.0)

⇒ In-bag-risk: 434,686.0 ; OOB risk (10-fold CV): 446,450.0 (previous lin. example with 1000 iter.)

- Feature importance (risk reduction over iter.)

  $\rightsquigarrow$ days_since_2011 most important

- Total effect for days_since_2011

  $\rightsquigarrow$ Combination of partial effects of linear BL and centered spline BL

MODEL-BASED BOOSTING - LINEAR EXAMPLE

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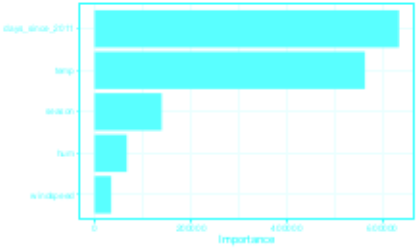
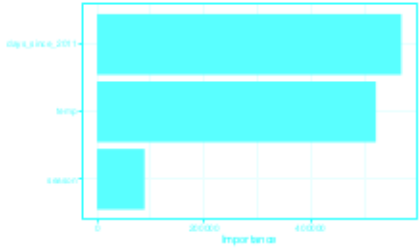


Overall risk: 434,686.0
OOB risk (10-fold CV): 446,450.0

20 iterations



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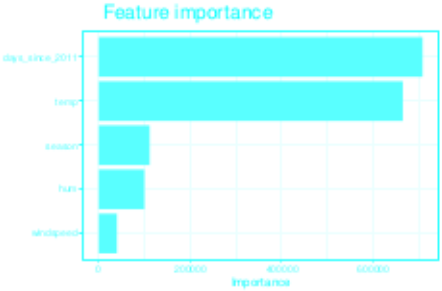

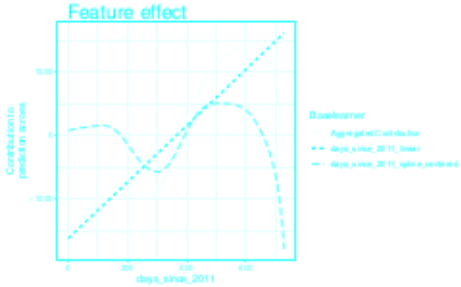

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- Total effect for `days_since_2011`
  ⇝ Combination of partial effects of linear BL and centered spline BL